

MRUGSEN GOPNARAYAN

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ABOUT ME

PhD student in Psychology at the University of Hamburg studying human and machine intelligence. I combine behavioral experiments, eye tracking, EEG, hierarchical Bayesian modeling, recurrent neural networks, and LLM evaluation to investigate social inference from choices and process-level cues. I am interested in foundation models of cognition, theory of mind in LLMs, and alignment between model representations and neural data.

SELECTED RESEARCH

Human–AI Preference Inference Evaluating how well LLMs perform social inference from human behavioral traces, with particular emphasis on whether they integrate information beyond choices, including response times and gaze.

Interactive Preference Inference in Bargaining Designed a repeated-interaction eye-tracking task with 75 dyads and developed a hierarchical Bayesian model of how sellers integrate choices, response times, and gaze to infer buyers' latent preferences.

Individual Differences in Value-Based Choice Investigated how anxiety and depression alter value-to-evidence translation using hierarchical drift-diffusion and prospect-theoretic models; co-developed the human-subject protocol and synchronized behavioral, EEG, and eye-tracking acquisition pipeline.

WORK EXPERIENCE

Full-Time Research Assistant

Decision Lab, Prof. Arjun Ramakrishnan

Aug 2021 – Feb 2023

Indian Institute of Technology, Kanpur

- Studied effects of trait anxiety and depression on decision-making.
- Built and maintained EEG/eye-tracking pipelines (LiveAmp, Pupil Capture, LSL).
- Collected synchronized behavioral, EEG, and eye-tracking datasets.

EDUCATION

University of Hamburg, Hamburg

PhD, Psychology. Thesis: How Humans and Machines Infer Preferences

Advisor: [Prof. Sebastian Gluth](#)

Mar 2023 – Present

Indian Institute of Science, Bangalore

Master of Science, Biology. Thesis: Modeling sentence reading time.

Advisor: [Prof. S. P. Arun](#)

Sep 2020 – Jul 2021

Indian Institute of Science, Bangalore

Bachelor of Science, Biology. Thesis: Evaluation of Cryo-EM structures.

Advisor: [Prof. N. Srinivasan](#)

Aug 2016 – Aug 2020

PUBLICATIONS

Mrugsen Nagsen Gopnarayan*, Liis Harjo, Jaan Aru, and Sebastian Gluth. 2026. Large language models infer hidden preferences from eye movements but fail to exploit response times. *PsyArXiv*. [PREPRINT](#). [\[link\]](#)

Mrugsen Nagsen Gopnarayan*, F. Sheng, M. L. Platt, and A. Ramakrishnan. 2026. Attenuation of value-to-evidence translation drives biased decision making in anxiety and depression. *bioRxiv*. [PREPRINT](#); [UNDER REVIEW](#). [\[link\]](#)

Mrugsen Nagsen Gopnarayan*, Sophie Bavard, Erik Stuchlý, and Sebastian Gluth. 2026. Humans integrate gaze and decision cues for inferring preferences in social interactions. *bioRxiv*, 2026.07.09.737460. [PREPRINT](#). [\[link\]](#)

Erik Stuchlý*, **Mrugsen Nagsen Gopnarayan**, and Sebastian Gluth. 2025. Neural dynamics of predicted decisions for similar and dissimilar others. *OSF*. [PREPRINT](#). [\[link\]](#)

Mrugsen Nagsen Gopnarayan*, Jaan Aru, and Sebastian Gluth. 2024. From DDMs to DNNs: Using process data and models of decision making to improve human-AI interactions. *Decision*. [\[link\]](#)

Mrugsen Nagsen Gopnarayan*, Deeksha R. Rathore, Fabio S. Bauer, Jasper Hillard, and Prerita Chawla. 2022. Differential representation of natural and man-made images in the human ventral visual stream. Neuro-match Conference 2022. [\[link\]](#)

Ashraya Ravikumar, **Mrugsen Nagsen Gopnarayan**, Sriram Subramaniam, and Narayanaswamy Srinivasan*. 2021. Comparison of side-chain dispersion in protein structures determined by cryo-EM and X-ray crystallography. *IUCrJ*. [\[link\]](#)

* Corresponding author

SKILLS

Programming: Python (PyTorch, TensorFlow, scikit-learn, NumPy), R, MATLAB, Bash, Git, $\text{\LaTeX}$

Computational Modeling: Hierarchical Bayesian models, DDMs, RNNs, HMMs, SVMs

AI Workflows: Psychologically grounded LLM evaluation and prompting, deep learning, Ollama

Experimentation: Human-subject and online studies, Pavlovia, Prolific, Pupil Labs, EEG

Statistics: SciPy, R, SPSS (t-tests, ANOVA, non-parametric tests)

Languages: English, Hindi, Marathi (Native), German (A2)

HONORS AND AWARDS

COSYNE 2024 Travel Award

- Received travel award for a high-rated abstract.

KVPY Fellowship

- Awarded for 5 consecutive years through a national-level exam and interview process (~100,000 applicants).

INSPIRE Fellowship

- Innovation in Science Pursuit for Inspired Research fellowship.

TEACHING EXPERIENCE

Course Follow-up Seminar: General Psychology (Instructor 2024, 2025)

- Led follow-up seminars to reinforce core concepts through discussion and short exercises.

BSE 662: Brain and Decision Making (Teaching Assistant)

- Mentored two student groups on project design, data collection, analysis, and presentation.
- Held weekly office hours to clarify course concepts and methods.

BSE 658A: Statistics for Modern Biology (Teaching Assistant)

- Graded weekly R-based assignments and provided feedback.
- Supported students with coding and statistical questions.

MENTORSHIP

BSc Students: Sophia, Cynthia, Mats, Ugur

MSc Student: Mario

REFERENCES

Prof. Sebastian Gluth

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Prof. Jaan Aru

Email: jaan.aru@ut.ee

Prof. Arjun Ramakrishnan

Email: arjunr@iitk.ac.in